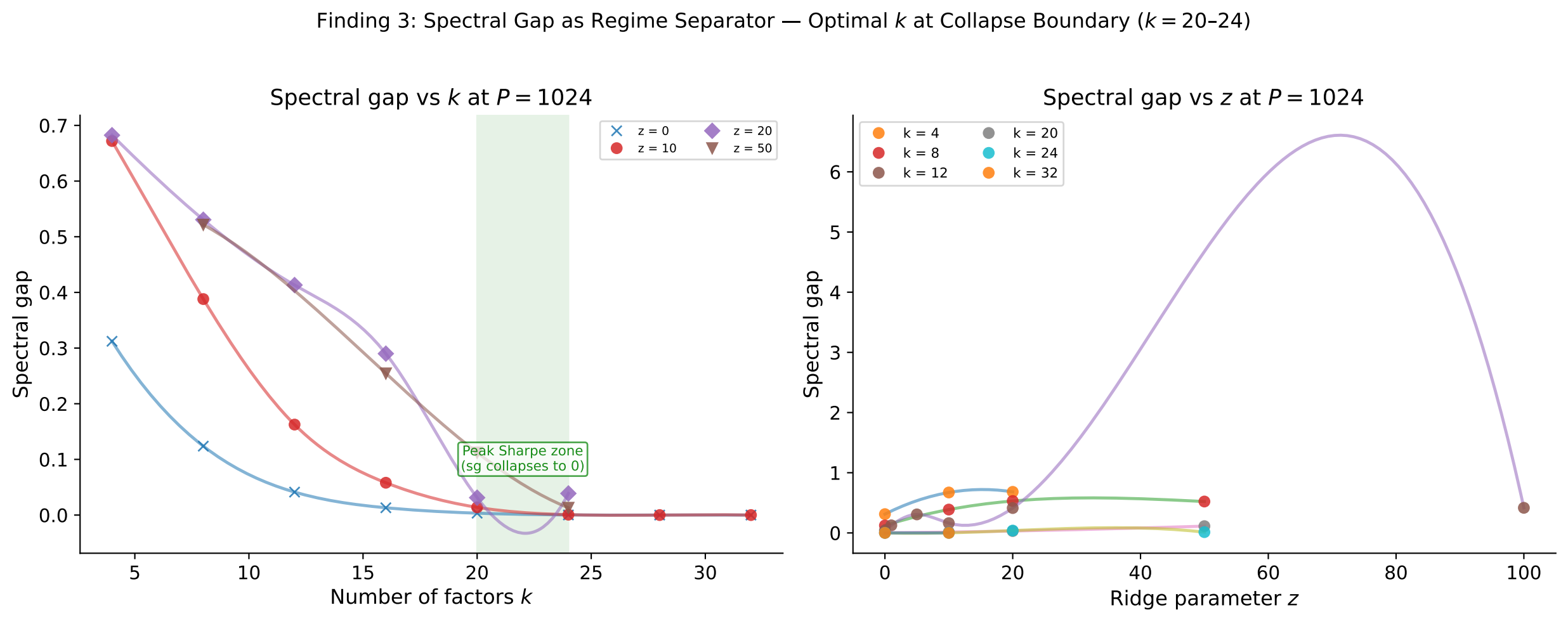

Finding 3: Spectral Gap as Regime Separator — Optimal k at Collapse Boundary ($k = 20$–24)
Spectral gap vs k at $P = 1024$
Spectral gap
Number of factors k
z = 0
z = 10
z = 20
z = 50
Peak Sharpe zone
(sg collapses to 0)
Spectral gap vs z at $P = 1024$
Spectral gap
Ridge parameter z
k = 4
k = 8
k = 12
k = 20
k = 24
k = 32